



LLMs Unplugged

Understand AI by building it yourself

Speaker notes

Open with the promise: understand AI by building one yourself, by hand. No computers, no maths background needed.

Acknowledgement of Country



Speaker notes

Acknowledge the Traditional Custodians of the land you're meeting on, and pay respects to Elders past and present. Keep it brief and sincere; say it in your own words.

activity

everyone stand up

Speaker notes

Quick energiser. Read each line and let people sit; it speeds up, and by "the last 5 minutes" almost everyone's down. Lands the point that this stuff is already everywhere. ~2 min, don't linger.

sit down if you
have **never** used ChatGPT

sit down if you
haven't used it in the last **month**

sit down if you
haven't used it in the last **week**

sit down if you
haven't used it in the last **day**

sit down if you
haven't used it in the last **hour**

sit down if you
haven't used it in the last **5 minutes**

What is this about?

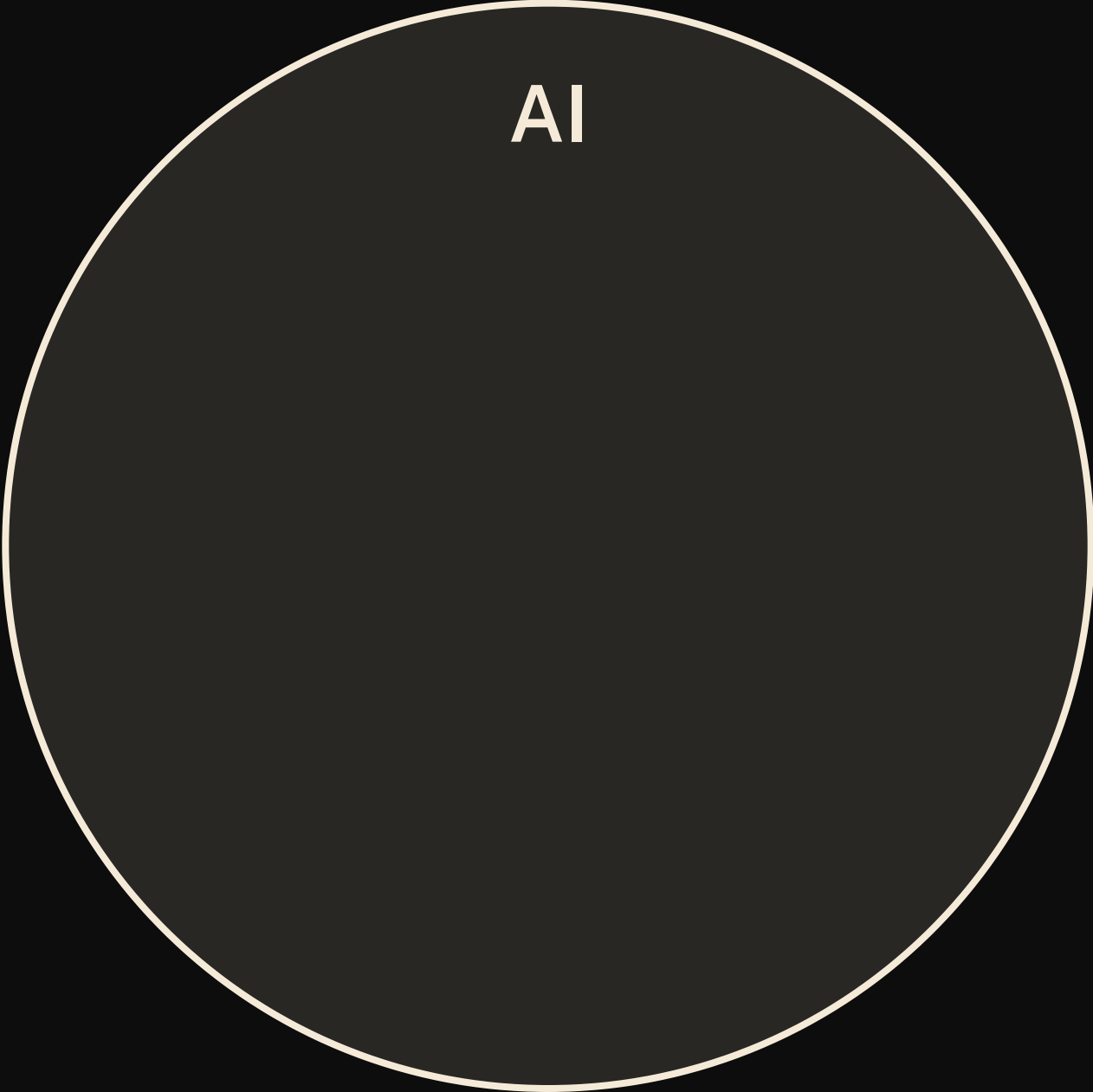
you'll **build** your own language model — from scratch — with just a kids book, pen & paper, and some dice rolling

you'll **learn** how language models work by spotting patterns in text to generate new text



Speaker notes

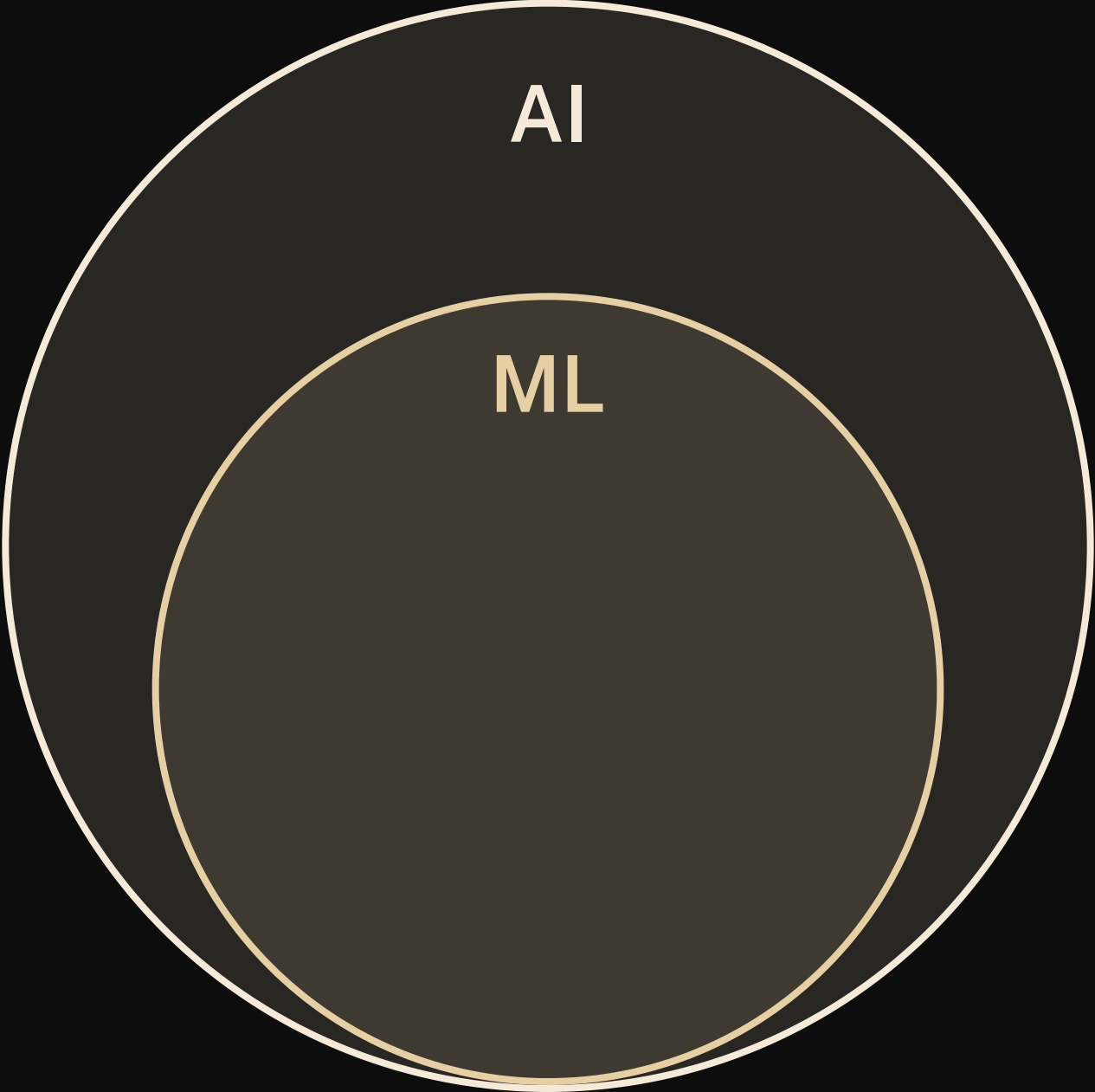
Frame the next stretch: build a model from a kids' book, pen, paper and dice, by spotting patterns. ~1 min, then move.



AI

Speaker notes

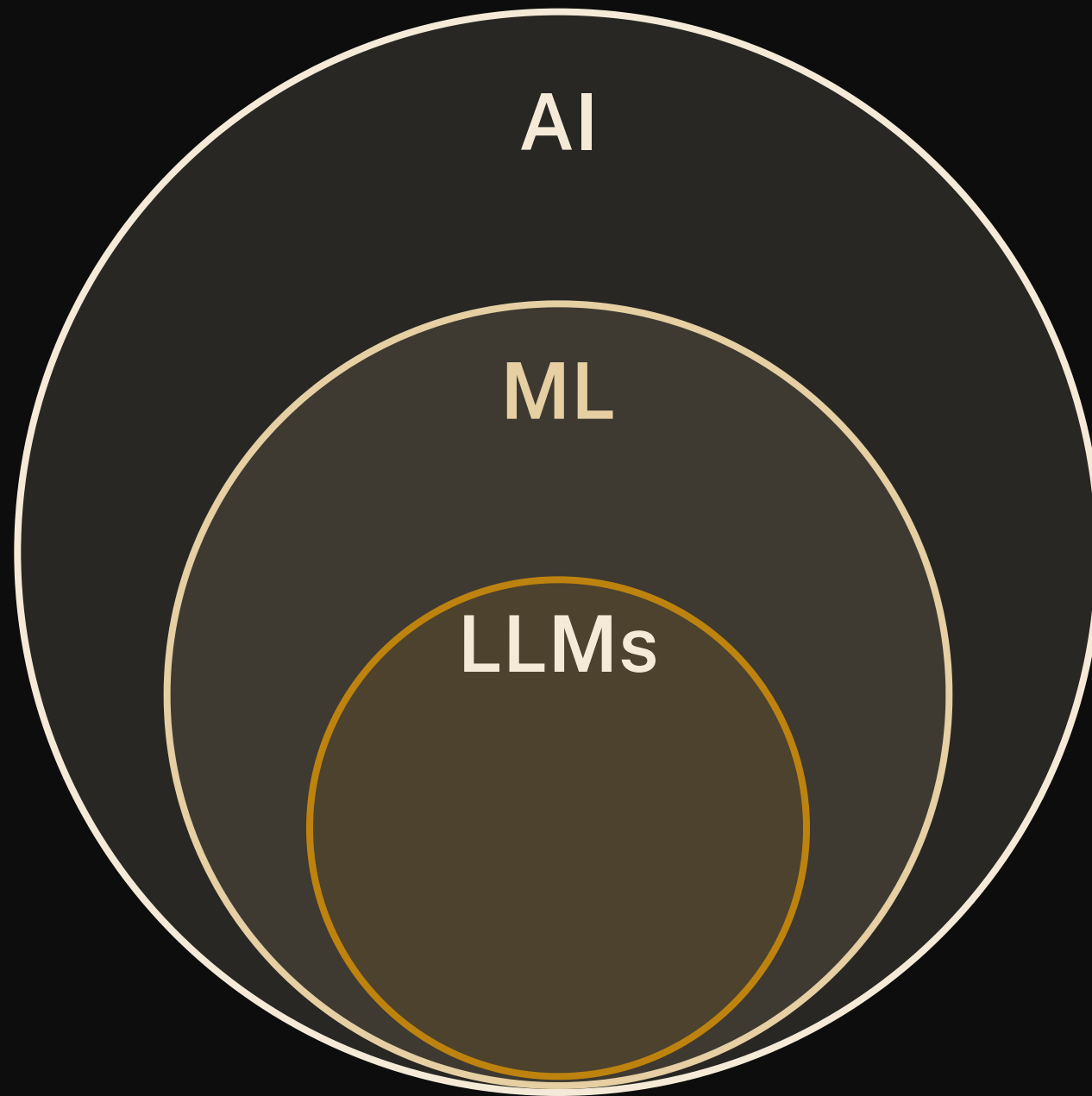
No text on this one --- talk over the build. AI: the umbrella term, any system doing things we'd call "smart" (chess engines, spam filters, route planners). Click --- ML: the subset that learns patterns from data instead of following hand-written rules; the model you'll build today lives here. Click --- LLMs: machine learning models trained on huge amounts of text to predict the next token --- exactly what you're about to do by hand, just scaled up. Click --- and the chatbots in the news (Claude, ChatGPT, Gemini, DeepSeek) are LLMs wrapped in an app. Same species as the thing you'll build with pen and dice.



A Venn diagram consisting of two concentric circles. The outer circle is dark gray and contains the text 'AI' at the top. The inner circle is a lighter shade of gray and contains the text 'ML' at the top. The inner circle is entirely contained within the outer circle, illustrating that Machine Learning is a subset of Artificial Intelligence.

AI

ML



A Venn diagram consisting of three concentric circles. The outermost circle is labeled 'AI'. Inside it is a circle labeled 'ML'. Inside the 'ML' circle is a smaller circle labeled 'LLMs'. The 'LLMs' circle contains a list of model names: Claude, ChatGPT, Gemini, and DeepSeek. The circles are shaded in a gradient from dark gray to a lighter brownish-gray.

AI

ML

LLMs

Claude
ChatGPT
Gemini
DeepSeek



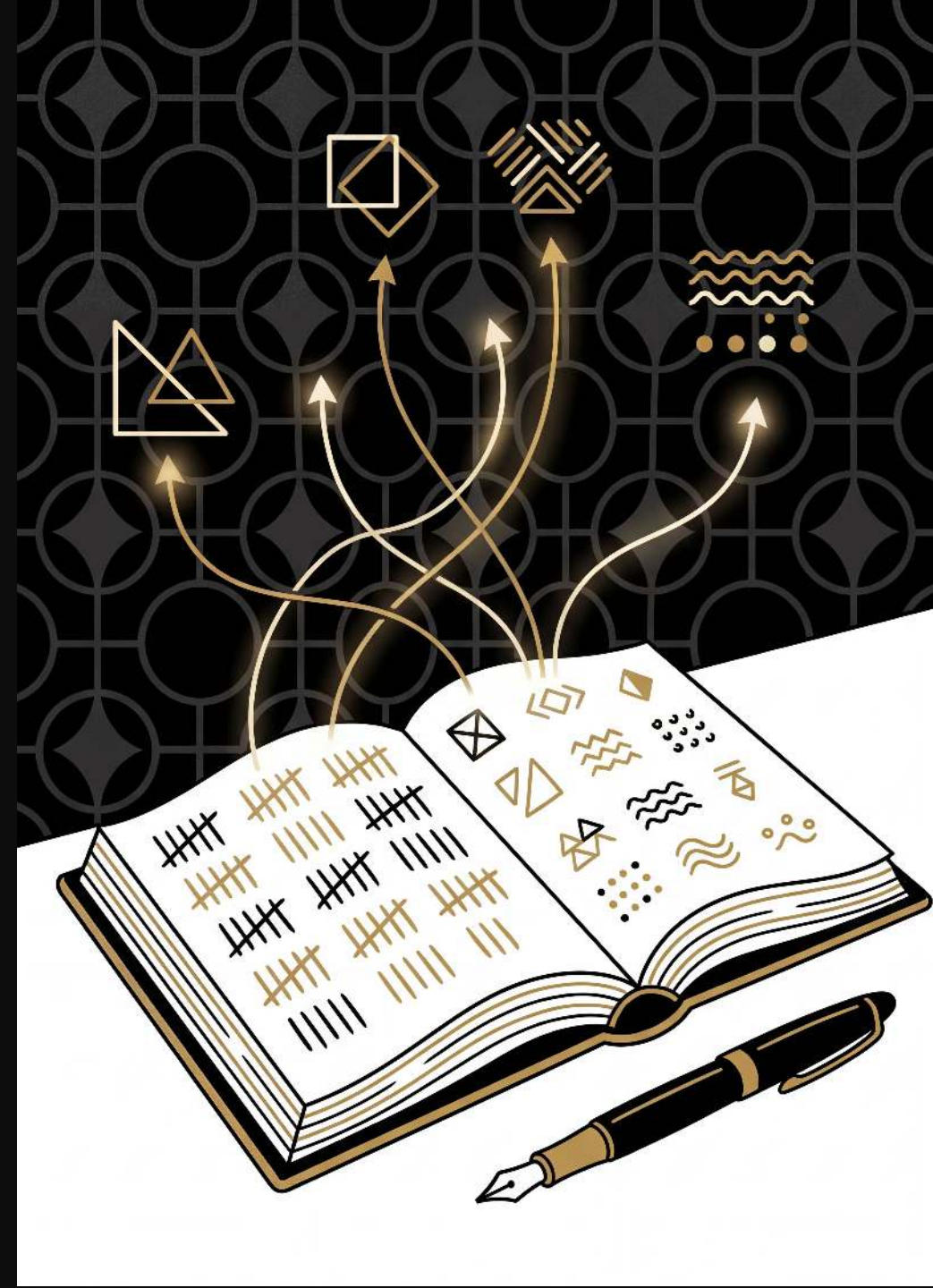
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Training



The recipe

walk through your text and tally up which tokens follow which in a grid



Speaker notes

1. **tidy up** your text: lowercase everything, treat words and single punctuation marks (. , ! ? ; :) as separate tokens
2. **set up** your grid: first token goes in both the row & column headers
3. **fill in** the grid: for each consecutive pair of tokens, add a tally mark in that cell (add new rows/columns as needed)

Example

“Run, Spot, run. See Spot run.”

after tidying up:

run , spot , run . see spot run
.



The empty grid

run , spot , run . see spot run .

Training: run → ,

run , spot , run . see spot run .

run ,					
run					
,					

Training: , → spot

run , spot , run . see spot run .

run , spot					
run					
,					
spot					

Training: spot → ,

run , spot , run . see spot run .

run , spot					
run					
,					
spot					

Speaker notes

, is already in our vocabulary — no new column needed!

Training: , → run

run , spot , run . see spot run .

run , spot					
run					
,					
spot					

Training: run → .

run , spot , run . see spot run .

	run	,	spot	.	
run					
,					
spot					
.					

Speaker notes

- . is a new token — punctuation gets its own row and column too

Complete model

run , spot , run . see spot run .

run		,	spot		see
run					
,					
spot					
.					
see					

Speaker notes

with the rest of the text tallied the same way, the grid is complete. run → . came up twice, so its cell has two marks — those counts are what make some next-words more likely than others.

Training

10:00

start

reset

+1

-1

Speaker notes

Hand out grids; groups tally their own text. 10 min, circulate --- the usual snag is that a token only gets a new row and column the first time it appears.

The *language* of language models

- model
- token
- vocabulary
- training



Speaker notes

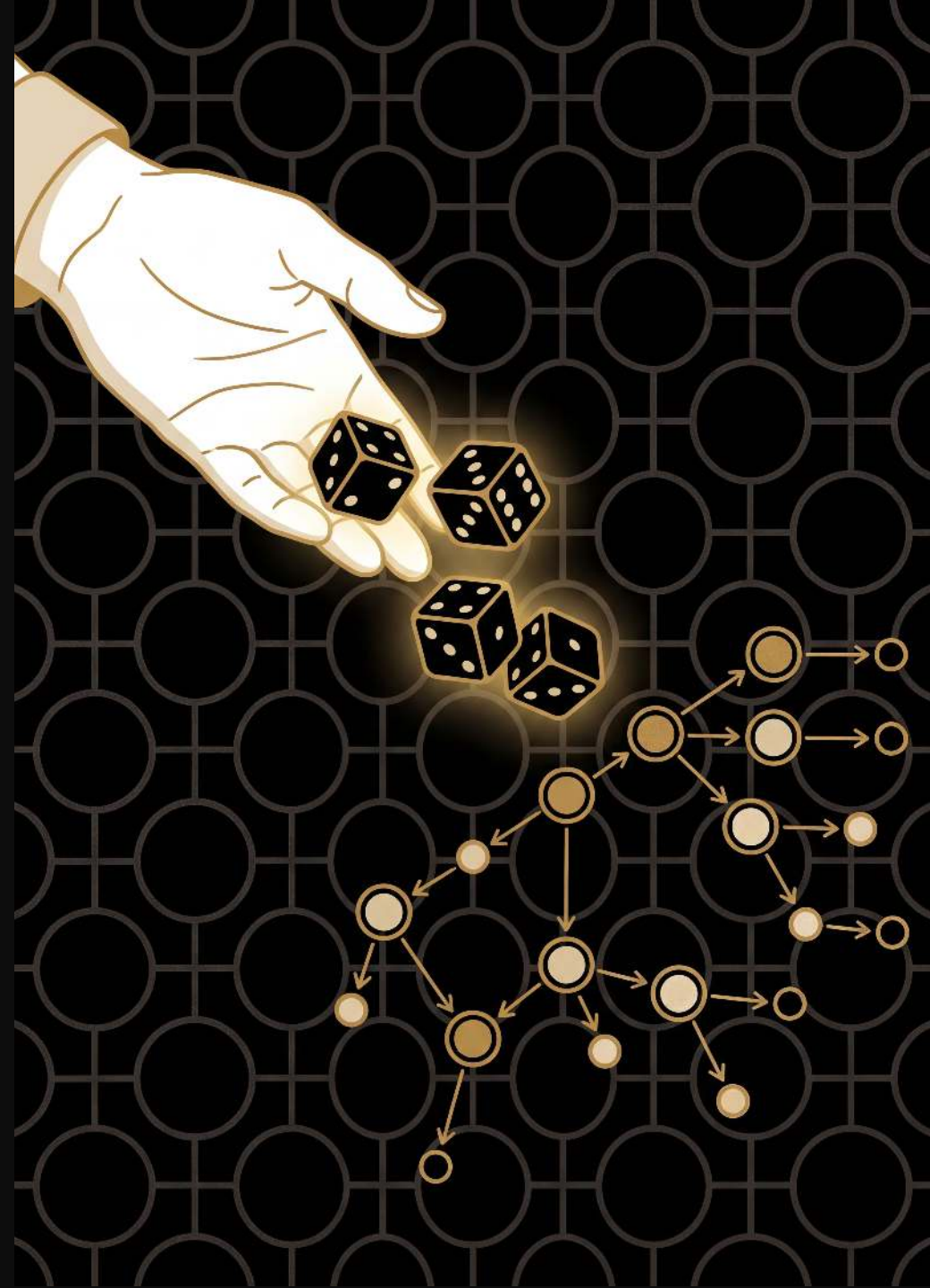
Gloss each: model = the thing that captures the patterns (your grid); token = one word or punctuation mark; vocabulary = all the distinct tokens it has seen; training = counting which token follows which.

Generation



The recipe

use your grid to generate new text, rolling dice
to choose each next word



Speaker notes

1. **choose** a starting word from your grid (any word --- not just the first one)
2. **look** at that word's row to find all possible next words and their counts
3. **roll dice** weighted by the counts to pick the next word
4. **write down** the chosen word and make it your new starting word
5. **repeat** from step 2

Generation: start with see

see

spot

one option — no roll needed

	run	,	spot	.	see
run					
,					
spot					
.					
see					

Speaker notes

any word will do as a seed ---we're starting from `see`, which is the last row, to show you don't have to begin at the top. Only one option here, so no roll.

Generation: from spot

see spot

2 options — roll the die!

	run	,	spot	.	see
run					
,					
spot					
.					
see					

Speaker notes

two options in this row (,` and `run`) with equal tallies ---highlight both, then roll

How the die chooses: **spot**

spot → ? roll a d10

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---

,

1 tally

run

1 tally

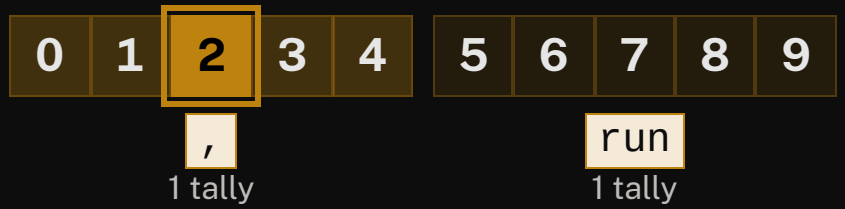
equal tallies → equal chances

Speaker notes

equal tallies, so each option gets an equal share of the ten d10 faces ---0-4 for one, 5-9 for the other. Pause here before rolling.

How the die chooses: **spot**

spot → ? roll a d10



equal tallies → equal chances

rolled **2** → ,

Speaker notes

now roll --- a 2 lands in the `` band

Generation: spot → ,

see spot ,

rolled 2 → ,

	run	,	spot	.	see
run					
,					
spot					
.					
see					

Speaker notes

the 2 landed in the `` band, so `` is our next word

Generation: from ,

see spot ,

2 options — roll the die!

	run	,	spot	.	see
run					
,					
spot					
.					
see					

Speaker notes

another two-option row (`spot` and `run`), still equal --- both highlighted, then roll

Generation: , → run

see spot , run

rolled 7 → run

	run	,	spot	.	see
run					
,					
spot					
.					
see					

Speaker notes

rolled a 7, which lands in the `run` band

Generation: from run

see

spot

,

run

2 options — roll the die!

	run	,	spot	.	see
run					
,					
spot					
.					
see					

Speaker notes

two options again ---but `` has two tallies to `,'s one, so this isn't a 50/50 roll

How the die chooses: **run**

run → ? roll a d10



more tallies → more faces → more likely

Speaker notes

` has twice the tallies, so it gets more of the faces (0-6) than ` (7-9). With a ten-sided die that's the closest split rather than exactly 2:1 --- the point is more tallies → more faces → more likely. Pause here before rolling.

How the die chooses: **run**

run → ? roll a d10



more tallies → more faces → more likely

rolled **3** → .

Speaker notes

now roll --- a 3 sits in the `` band

Generation:

run

 →

.

see

spot

 ,

run

.

rolled **3** →

.

	run	,	spot	.	see
run					
,					
spot					
.					
see					

Speaker notes

the 3 sits in the `` band, so we pick ``

Generation: from .

see

spot

,

run

.

see

one option — no roll needed

	run	,	spot	.	see
run					
,					
spot					
.					
see					

Speaker notes

only one option (`see`) ---no roll. Note we keep rolling straight past the full stop; the model has no idea a sentence just ended.

Generation: back to see

see spot , run . see

one option — no roll needed

	run	,	spot	.	see
run					
,					
spot					
.					
see					

Speaker notes

and so on --- we've looped back to where we started. Keep going for as long as you like; you decide when to stop.

Generated text

“see spot, run. see”

a new sentence — not in the training data!



Generation

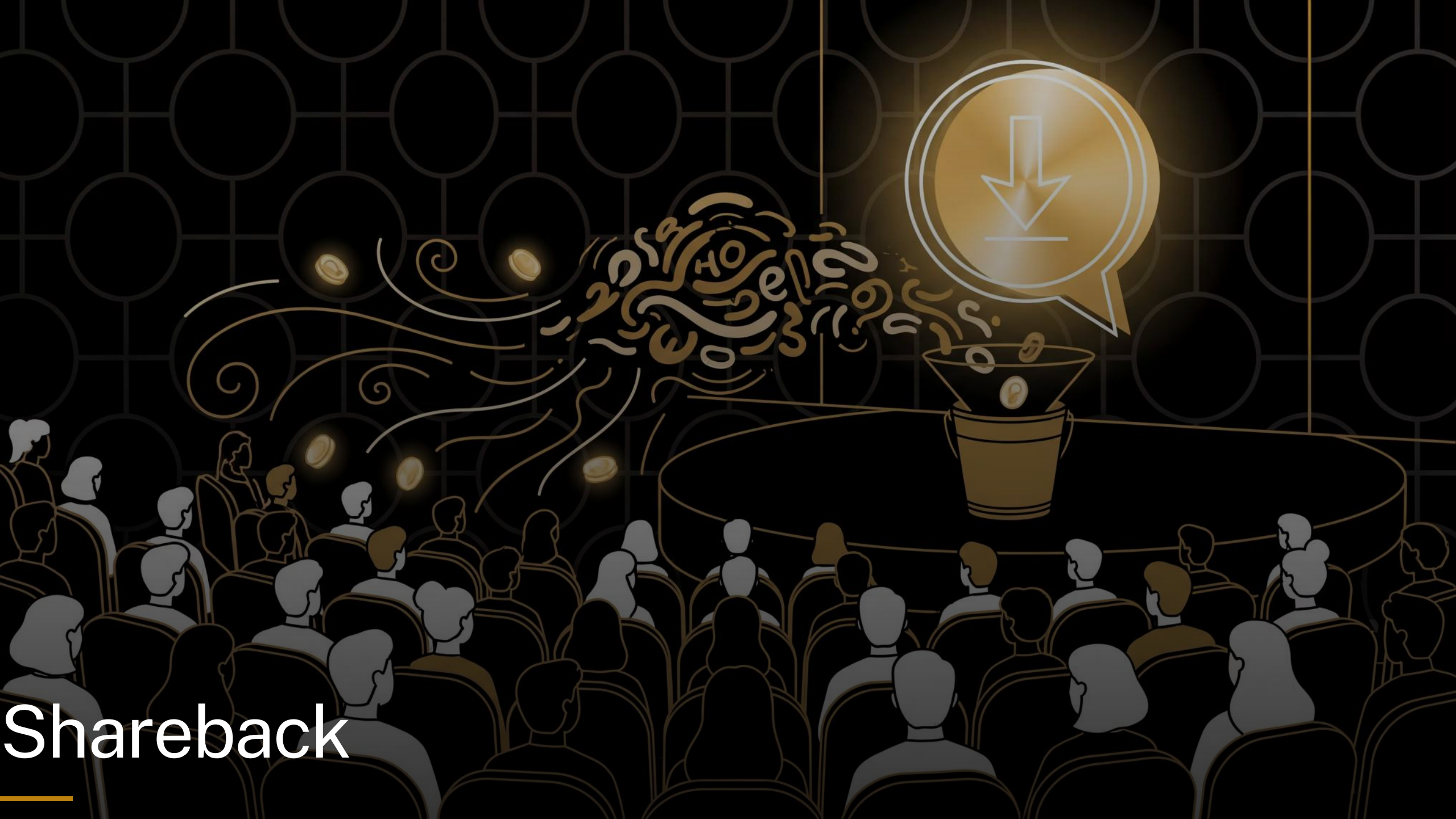
10:00

start reset +1 -1

Speaker notes

Groups generate from their grid, rolling dice on each choice. 10 min, circulate. Where a row has only one option, no roll needed.

Shareback



Speaker notes

Take a couple of groups' generated sentences; celebrate the weird ones. ~5 min.

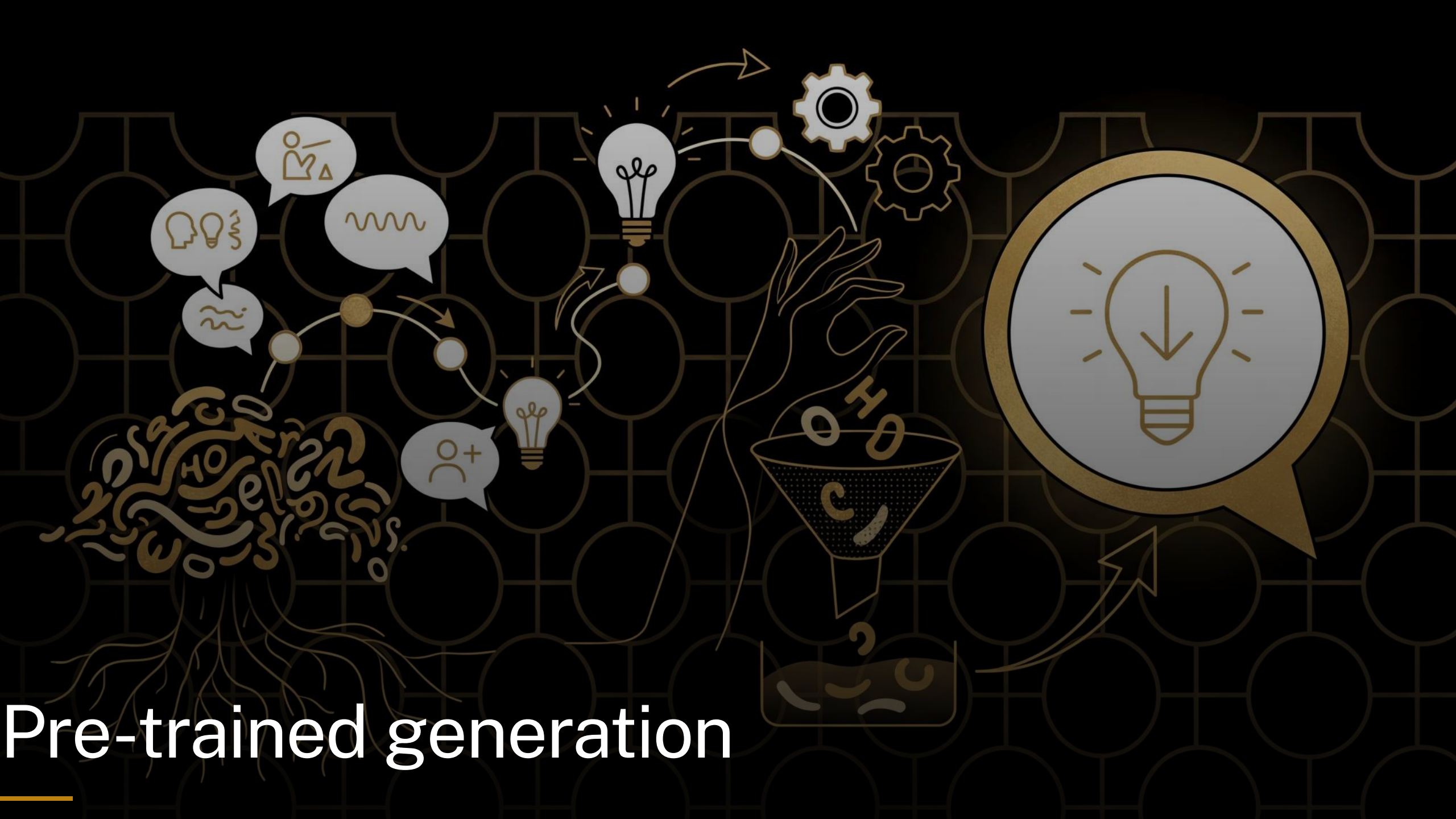


The *language* of language models

- prompt
- response/completion
- context window

Speaker notes

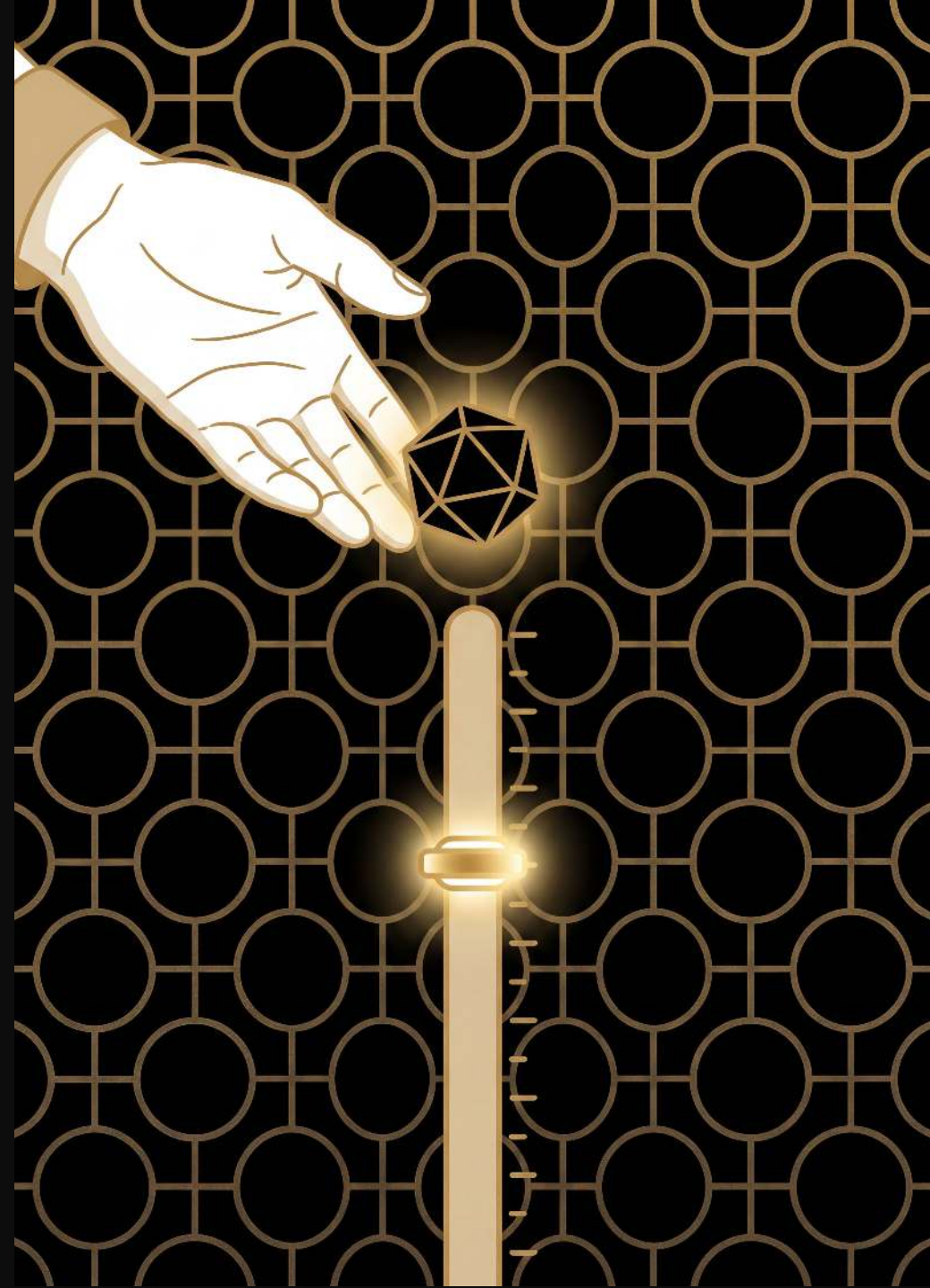
Gloss each: prompt = the starting text you give it; response/completion = the text it generates back; context window = how much it can "see" at once (here, just the current word).



Pre-trained generation

The recipe

use the booklet to generate new text, rolling
one or two d10s to choose each next word



Speaker notes

1. **choose** a starting word --- any bold word in the booklet
2. **look up** that word's entry to see possible next words and their thresholds
3. **roll** your d10(s) and scan the thresholds to find the next word
4. **write down** the chosen word and make it your new starting word
5. **repeat** from step 2

Pre-trained: start with **the**

the **cat**

the ♦♦ **49|cat** 79|dog 99|hat **rolled 27**

cat ♦ 5|sat 9|ran

sat □

□ **the**

ran □

dog ♦ 6|sat 9|ran

hat ♦ 4|sat 9|ran

Speaker notes

a bigger model — the is so common it needs **two** dice (roll two d10s for 00 to 99)

Pre-trained: from cat

the cat sat

the ♦♦ 49|cat 79|dog 99|hat

cat ♦ 5|sat 9|ran **rolled** 4

sat .

. **the**

ran .

dog ♦ 6|sat 9|ran

hat ♦ 4|sat 9|ran

Speaker notes

cat is rarer than the — just **one** die, roll a d10

Pre-trained: from **sat**

the cat sat .

the ♦♦ 49|cat 79|dog 99|hat

cat ♦ 5|sat 9|ran

sat .

. the

ran .

dog ♦ 6|sat 9|ran

hat ♦ 4|sat 9|ran

Speaker notes

only one option — no dice roll needed

Pre-trained: from

the cat sat  

the ♦♦ 49|cat 79|dog 99|hat

cat ♦ 5|sat 9|ran

sat 

 the

ran 

dog ♦ 6|sat 9|ran

hat ♦ 4|sat 9|ran

Speaker notes

only one option — no dice roll needed

Pre-trained: from **the** again

the cat sat . **the** dog

the ♦♦ 49|cat **79|dog** 99|hat **rolled 63**

cat ♦ 5|sat 9|ran

sat □

□ **the**

ran □

dog ♦ 6|sat 9|ran

hat ♦ 4|sat 9|ran

Speaker notes

two dice again — 63 is past cat's 49, so we land on dog

Pre-trained: from **dog**

the cat sat . the **dog**

the ♦♦ 49|cat 79|dog 99|hat

cat ♦ 5|sat 9|ran

sat □

□ the

ran □

dog ♦ 6|sat 9|ran

hat ♦ 4|sat 9|ran

Speaker notes

and so on...

Pre-trained generation

8:00

start

reset

+1

-1

Speaker notes

Same as before but from the booklet --- a bigger model, one or two d10s. 8 min, circulate.

Shareback

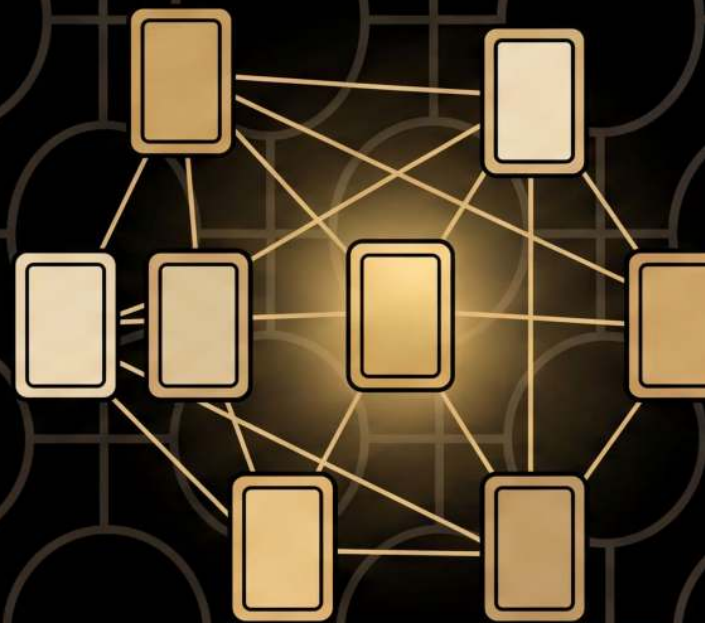


Speaker notes

Compare with the hand-built model: a bigger training text gives more varied, more natural output. Invite ~5 groups to share back rather than the whole room. ~5 min.

The *language* of language models

- pre-training
- foundation model



Speaker notes

Gloss each: pre-training = the big training run done ahead of time on huge amounts of text; foundation model = the general-purpose model that run produces, which other things are built on.

The background is a dark gray grid of circles. In the center, there is a cluster of approximately 15 glowing, rounded rectangular shapes. These shapes are arranged in a roughly circular pattern, with some overlapping. They have a yellowish-gold glow and a dark outline. The text 'Scaling up' is located in the bottom left corner, with a small orange horizontal line underneath it.

Scaling up

Speaker notes

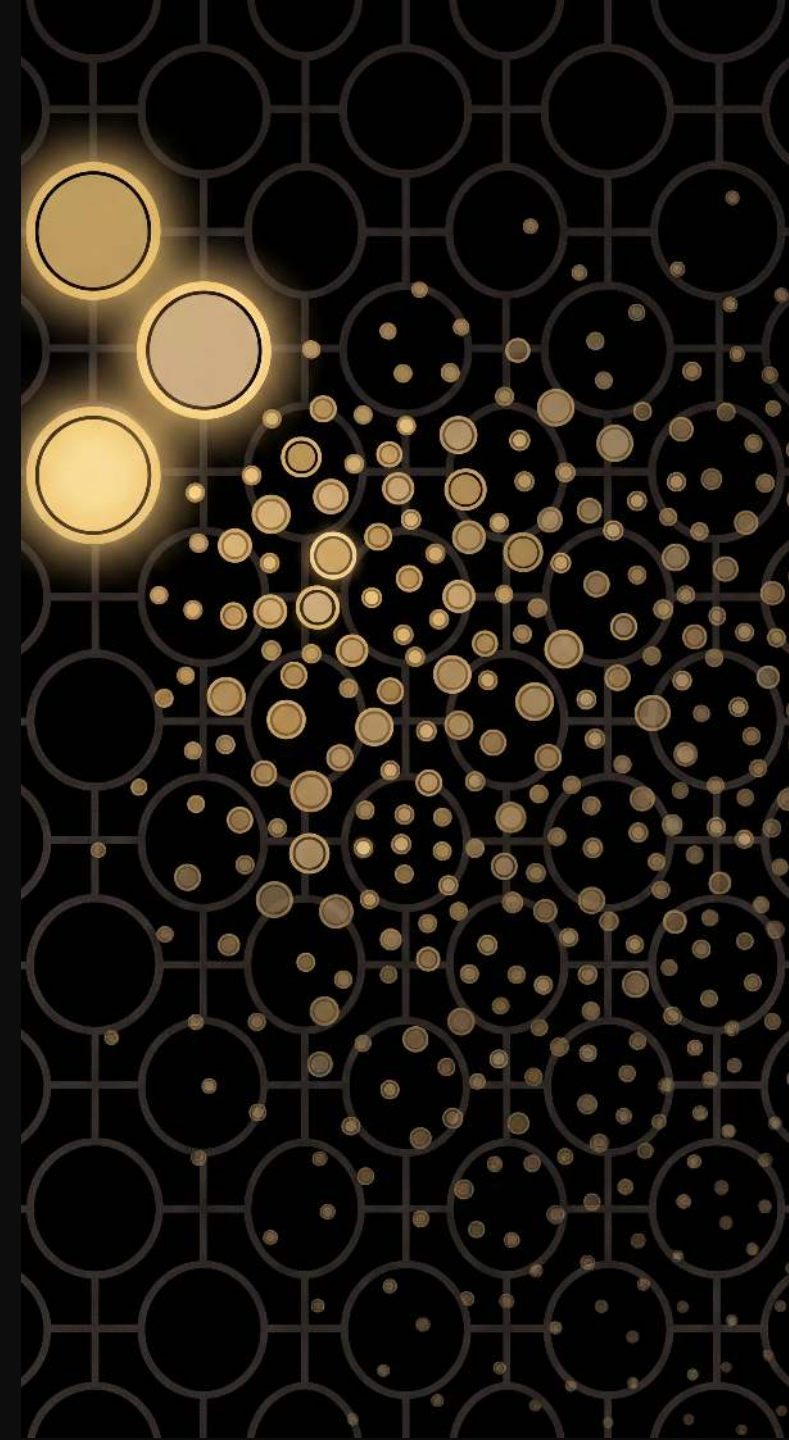
The reassurance beat: what they built isn't a cartoon of an LLM, it's the real mechanism at human scale --- look at the context, weigh every possible next word, roll the dice, write it down, repeat. So what's actually different? More context, attention, billions of numbers instead of tallies, and a second round of training. Each is a diagram coming up. ~8 min, keep it moving.

More context

your grid run , spot , run ?

Commercial LLMs ...hundreds of pages before this ... run , spot , run ?

your grid sees **one word**; Claude sees **hundreds of thousands**



Speaker notes

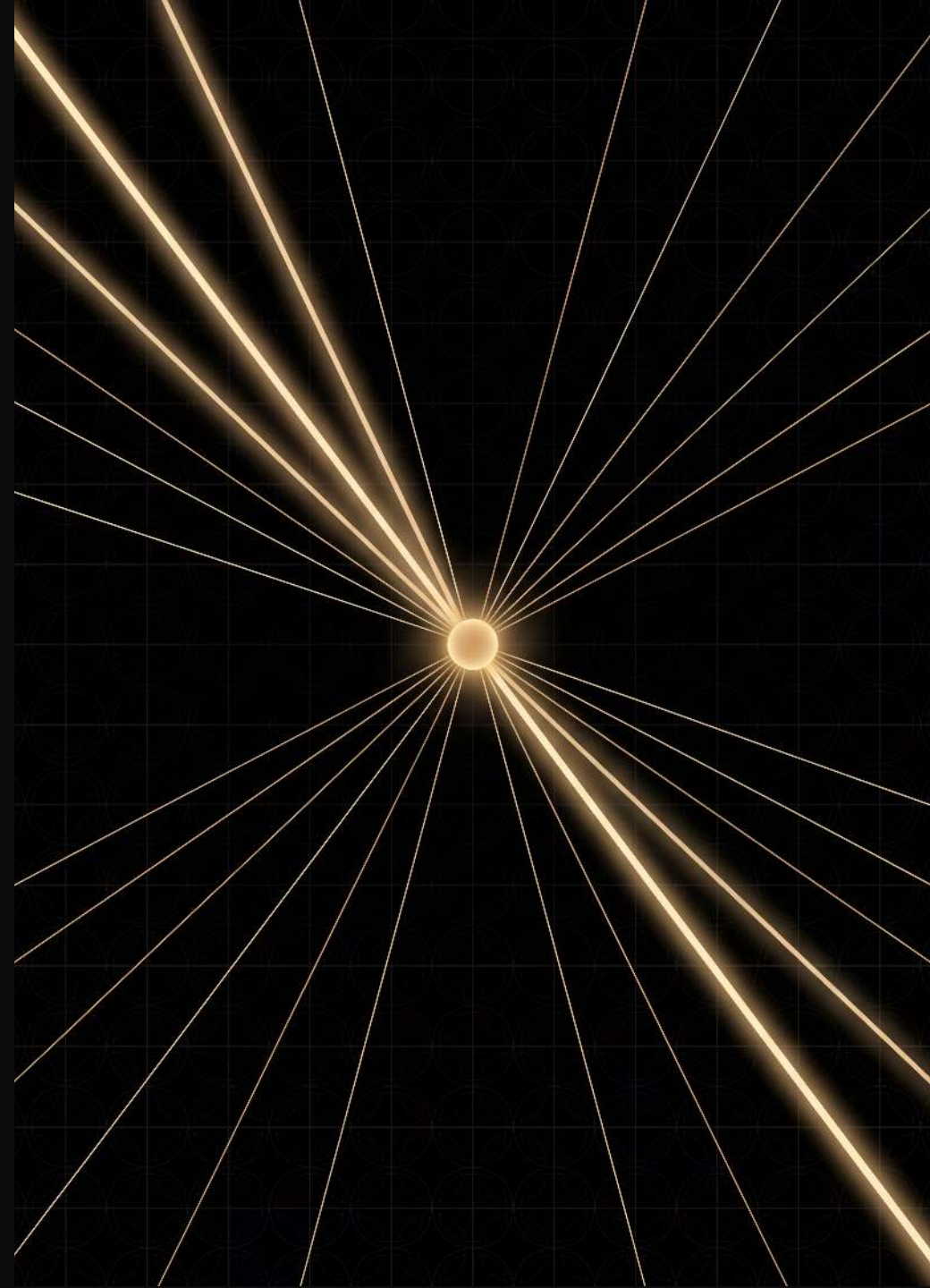
This is the "context window". One word of context is why their generated text wanders ---it forgets where it was going one word ago. Real models hold a few novels' worth in view for every single prediction.

Attention: learning where to look

the wifi **password** is **swordfish** ... the **password**
is ?

the next word is *swordfish* — attention finds where **password** last appeared and copies what came after

attention lets every earlier word vote on what comes next, and the model *learns* the weights — no grid could ever be that big



Speaker notes

The transformer's central trick, and the answer to the blown-up grid: instead of a row for every possible context, each prediction looks back over the whole context and weighs which words matter. Here it's copying outright --- the model spots the earlier "the password is swordfish" and repeats what followed. That look-back-and-copy move is an induction head, and the Induction Heads lesson on the website is exactly this pattern --- "find the last time I saw this word, copy what came next" --- run with pen and paper.

Tallies become knobs

your grid

run

.



a transformer

0.31

-1.20

0.08

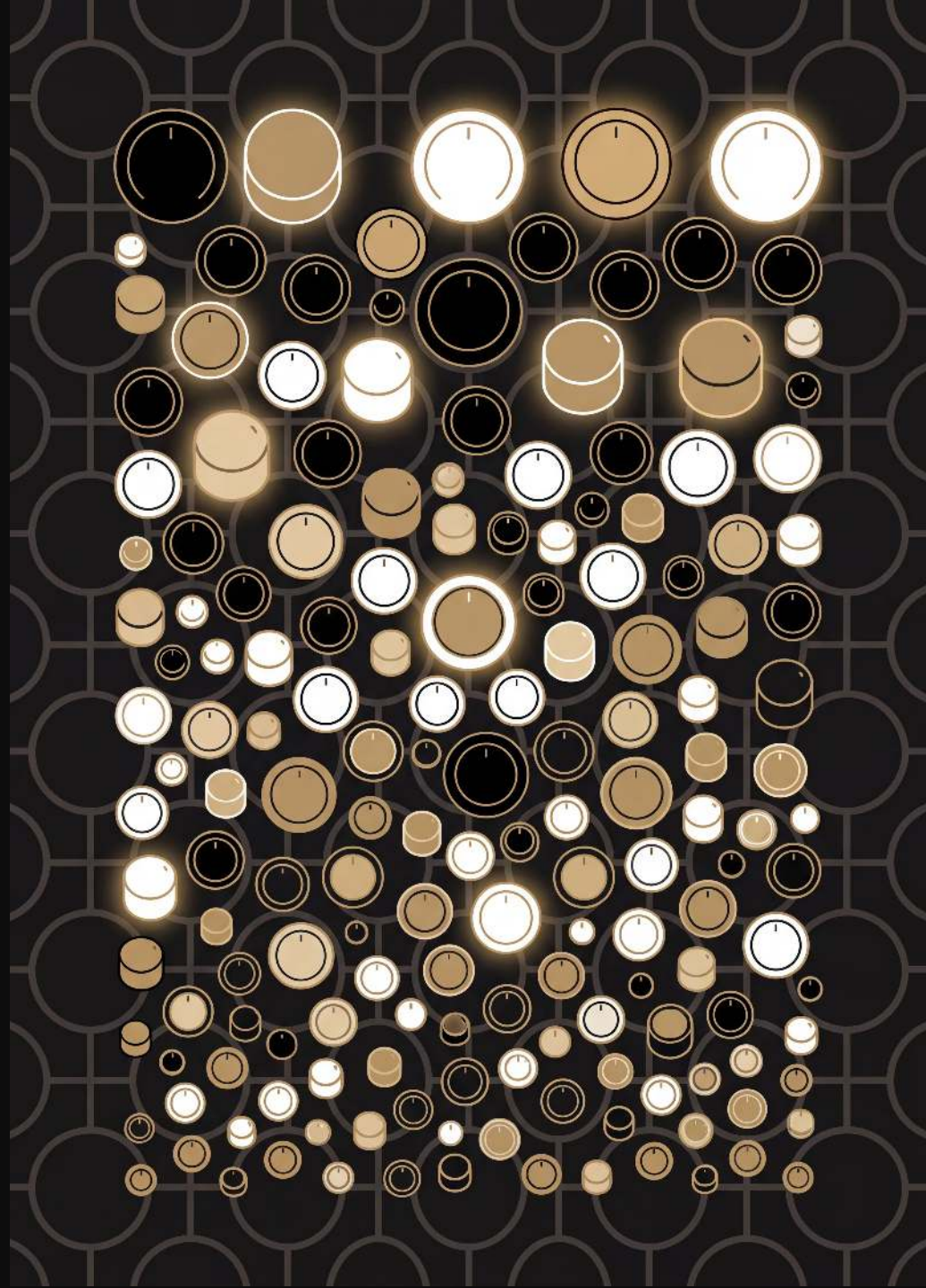
0.94

-0.55

1.30

× billions

your grid stores each pattern as **tally marks**
you can count; a transformer spreads the
same patterns across **billions of tunable**
numbers — knobs nudged during training, so
similar words like **dog** and **cat** come to
share them



Speaker notes

Your grid stores patterns as tally marks you can point to. A transformer spreads them across billions of numbers (its parameters), each nudged a tiny amount after every wrong guess in training. That spreading is why it generalises --- "dog" and "cat" end up sharing patterns. The cost: you can no longer point to where any one pattern lives.

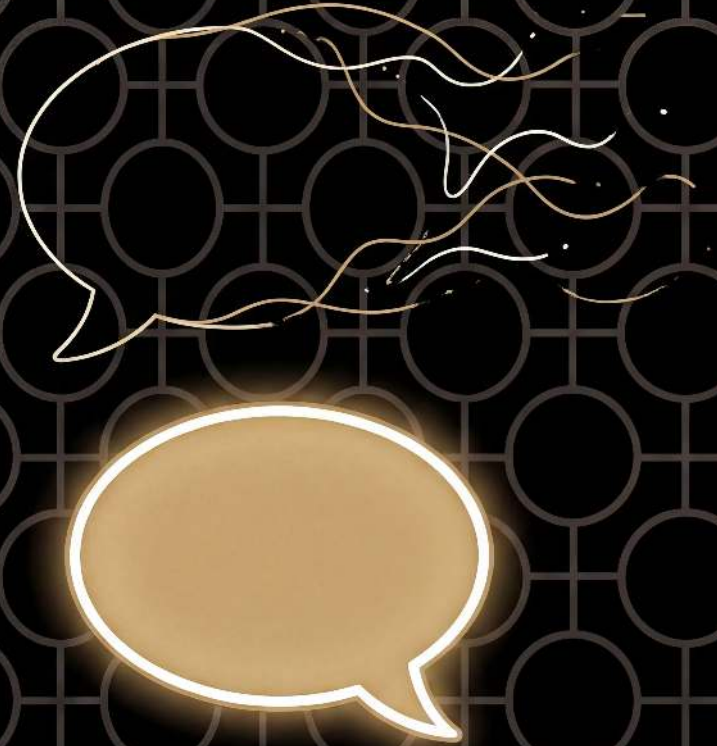
From continuing to answering

prompt: "what is the capital of France?"

a base model "What is the capital of Germany? What is the largest..."

+ post-training "Paris."

everything you built only **continues** text —
post-training on example conversations is
what turns a *continuer* into an *assistant*



Speaker notes

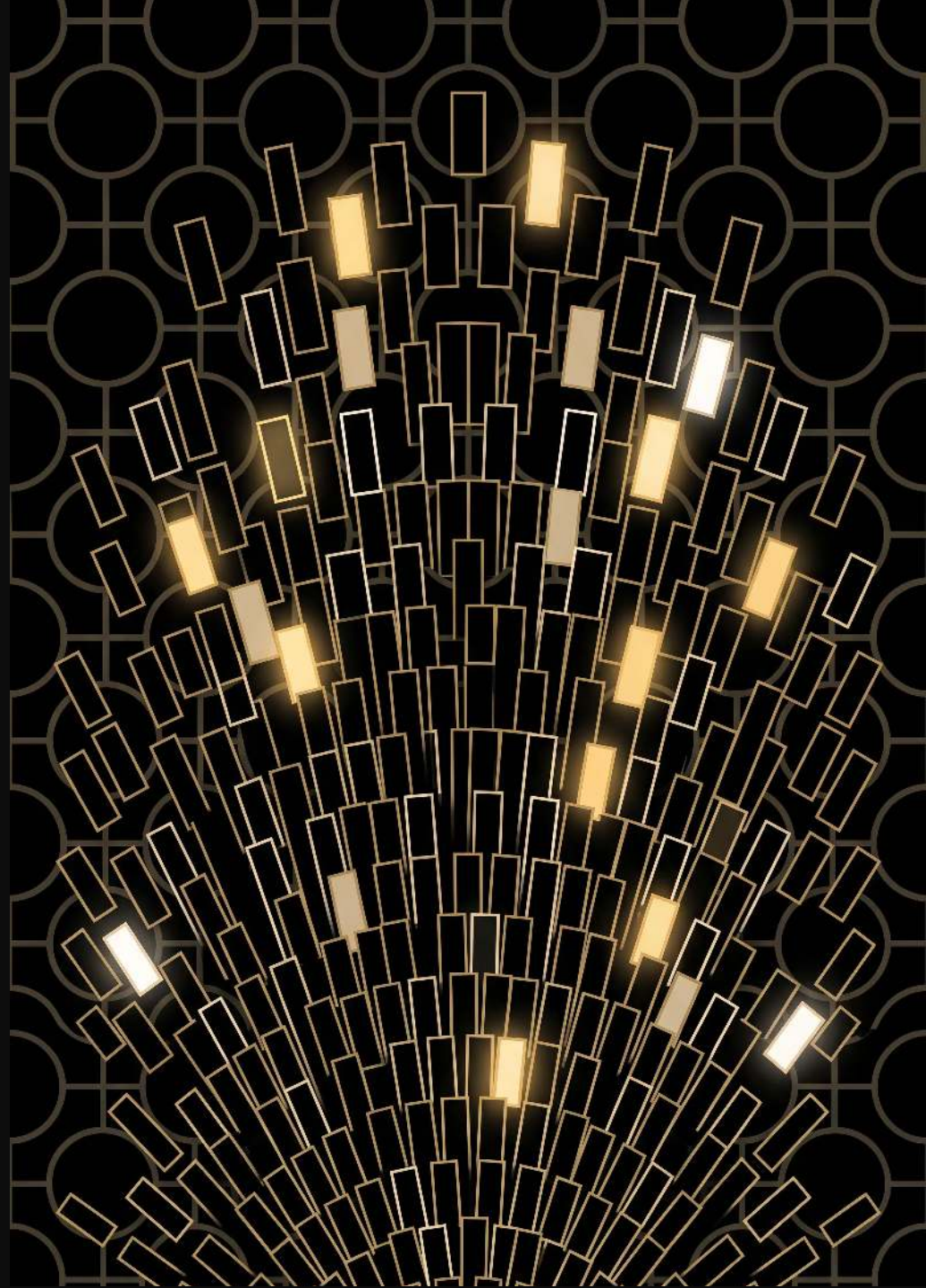
A pretrained model is a text-continuer, not an assistant --- prompt it with a question and a plausible continuation is more questions. The chat behaviour comes from a second, much smaller phase of training on example conversations plus feedback on which answers people preferred. Same predict-the-next-word mechanism underneath; different data, different habits on top.

Scale is a hell of a drug

your model trained on a few hundred words — a few dozen tallies

frontier LLM trained on tens of trillions of words — hundreds of billions of numbers

the same loop you ran by hand — now with **billions of times** the text and **billions of times** the numbers



Speaker notes

Put the two scales side by side. Training data: a page or two of text versus tens of trillions of words (the open figures land around 30 trillion --- Qwen3 cites ~36T, GLM-5.2 ~28.5T --- but the number keeps climbing, so keep it as "tens of trillions"). Parameters: a few dozen tally marks versus hundreds of billions of tunable numbers (frontier dense models run into the hundreds of billions; the big mixture-of-experts ones past a trillion). Two ballparks to make it physical. Paper: at the booklet's 1cm^2 per parameter, a frontier model's ~1.5 trillion parameters would need a sheet of about 150 km^2 --- a square roughly 12 km on a side, about 100 ANU Acton campuses, or twenty-odd Lake Burley Griffins (your grid was a postcard). Time: if hand-training 100 tokens took you ~10 minutes, 30 trillion tokens at that rate is ~5.7 million years --- you'd have had to start about when our ancestors split from the chimpanzees to be finishing now. The punchline isn't any one figure --- it's that the mechanism didn't change, only the dosage.

still just **tokens in** → **tokens out**

Speaker notes

Every reply from Claude or ChatGPT is sampled one word at a time, exactly like their dice rolls --- that's why "regenerate" gives a different answer. Bigger context, learned where to look, billions of numbers, a politeness pass --- but the loop never changed.



Questions

what new questions do you have about large language models?

What next?

how will this change the way you think about
and use LLMs in the future?





Australian
National
University

School of
Cybernetics

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